



Jadavpur
University

UNIFORM DISPERSION OF OBLIVIOUS MOBILE ROBOTS ON ALTERNATIVE NODES OF AN INFINITE GRID

Sudip Naskar, Subhadip Sinha, Subhas Chandra Shit and Sruti Gan Chaudhuri
Department of Information Technology, Jadavpur University, Kolkata, India

sruti.ganchaudhuri@jadavpuruniversity.in

+919836045650

Kolkata, West Bengal

Objective

- ❖ **Objective:** Develop a distributed algorithm to disperse autonomous, identical mobile robots across an infinite 2D grid. The algorithm operates using full visibility, semi-synchronous movement (SSYNC), and axis agreement among the robots.
- ❖ **Goal State:** Robots uniformly occupy alternate nodes, maximizing regional coverage with the minimal number of units.



Introduction

- ❖ **Research Focus:** Investigates decentralized coordination of autonomous, homogeneous, identical, and oblivious (stateless) mobile robots on an infinite 2D grid.
- ❖ **The Problem:** Coordinating scattered robots to autonomously form a specific geometric configuration occupying alternate nodes equidistantly without central control or explicit communication.
- ❖ **Approach:** The system uses a deterministic distributed algorithm operating on a **Look-Compute-Move** cycle under a semi-synchronous scheduler. Robots dynamically compute the boundary limits (Y_{MAX} and X_{MIN}) of the workspace. They then independently execute specialized movement logic divided into 5 distinct spatial configurations (*Case ψ_1 to Case ψ_5*), using a strict directional priority order (West > East > South > North) to navigate row-by-row into empty alternate nodes while mathematically preventing deadlocks or collisions.



System Framework

- ❖ **Robot Set:** Consists of n autonomous mobile robots, modeled as points on an infinite grid ($R = \{r_1, r_2, \dots, r_n\}$).
- ❖ **Target Configuration:** Robots aim for grid_final, where they occupy alternate nodes.
- ❖ **Grid Dimensions:** Based on the number of robots n , the system calculates the maximum robots per row/column r_c and the total grid span (d) including empty nodes:
 - $r_c = \lceil \sqrt{n} \rceil$
 - $d = (r_c \times 2) - 1$
- ❖ **Logical Mapping:** Each robot r_i at (x, y) calculates a row index j relative to the northernmost boundary (Y_{MAX}):
 - $j = Y_{MAX} - y + 1$ (where $j=1$ is the North-most row).
- ❖ **Robot Characteristics:**
 - ❖ **Autonomous:** Operate independently without central control or external servers.
 - ❖ **Anonymous & Homogeneous:** Indistinguishable from one another with no unique IDs.
 - ❖ **Oblivious:** Stateless with no memory of past actions; decisions depend strictly on the current observation.
 - ❖ **No Communication:** No explicit message passing; robots rely entirely on sensing (e.g., 360-degree vision) to detect others.
 - ❖ **Unlimited Visibility:** Each robot can observe the position of every other robot across the entire grid.
- ❖ **Timing Model (SSYNC):** Operates on a *Semi-Synchronous Scheduler* divided into discrete **Look-Compute-Move** cycles.
 - A random subset of robots is activated per cycle.
 - The scheduler guarantees snapshots are never taken while another robot is in motion.
- ❖ **Coordinate System:** All robots share a common global coordinate system, agreeing on a shared origin $(0, 0)$ and the exact orientation of the X and Y axes.

LOOK

The activated robot scans the infinite 2D grid using its 360-degree sensing capability. It records the exact positions of all other n robots relative to its own coordinate system.



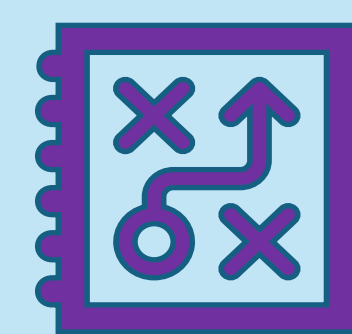
MOVE

The robot calculates its target destination on the grid to assist the dispersion algorithm and physically moves to the new node.



COMPUTE

Using the snapshot data, the robot evaluates the structural formulas (calculating r_c , d , and its logical row index j). Because it is oblivious, this computation depends strictly on the current snapshot, ignoring past actions.



Conclusion

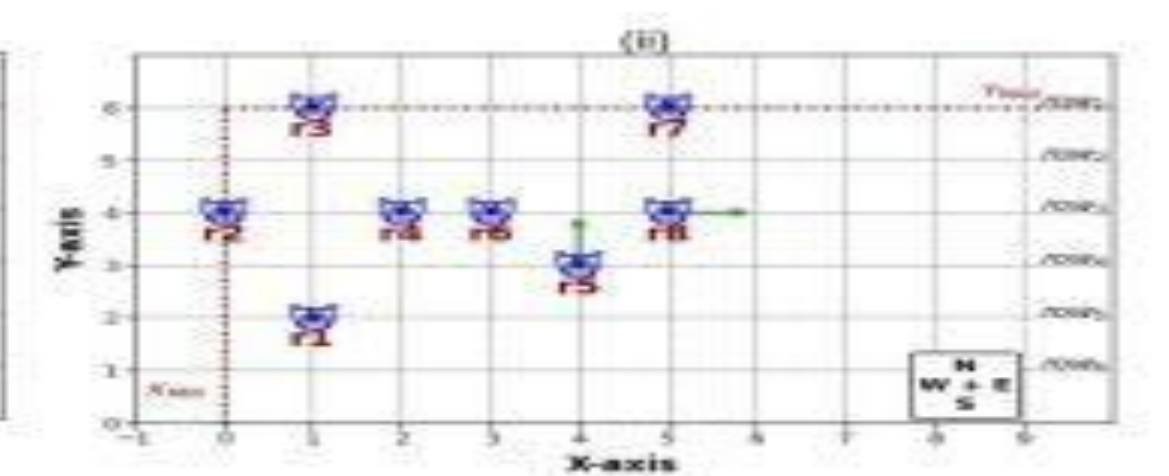
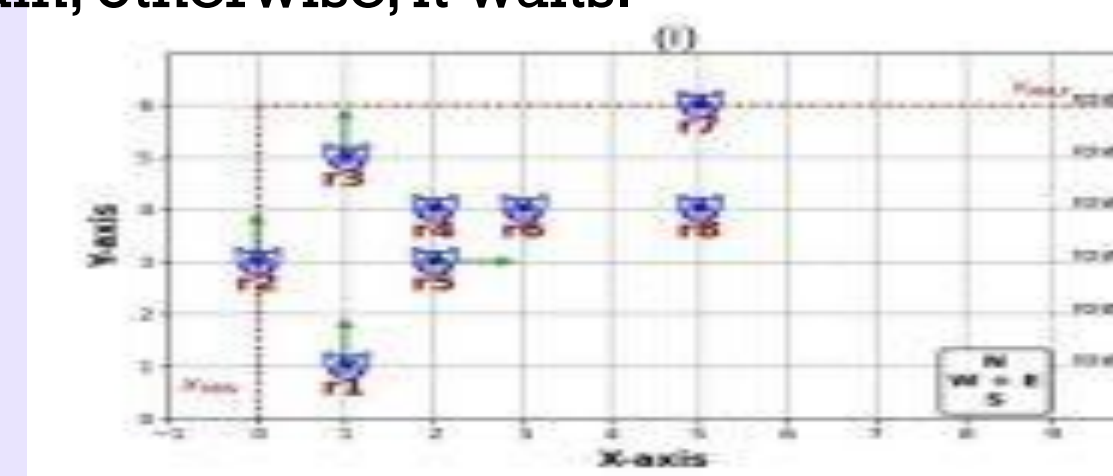
- ❖ **Impact:** Spacing memoryless robots on alternate nodes cuts hardware costs and uses up to 50% fewer units for full area coverage in spraying or sensing tasks.
- ❖ **Theorem:** We prove that autonomous, oblivious, identical robots will successfully disperse on alternate grid nodes in finite activation cycles from any arbitrary setup under an SSYNC scheduler.
- ❖ **Guarantees:** The deterministic algorithm mathematically ensures finite-time convergence, absolute collision avoidance, and a completely deadlock-free execution.
- ❖ **Future Scope:** Future work will adapt this swarm algorithm to operate under limited visibility constraints and focus on minimizing the total travel distance of the robots.

Movement Logic

- ❖ **Path Priority:** When multiple movement options exist, ties are resolved using strict directional priority: **West > East > South > North**.
- ❖ **Behavioral Configurations:** Each robot executes one of five distinct behavioral cases (ψ_1 to ψ_5) based on its current position:

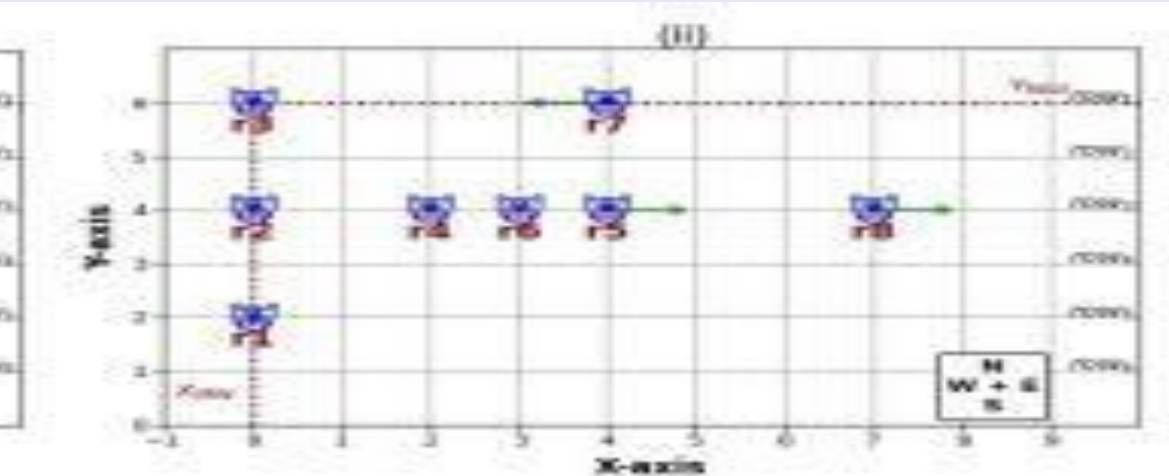
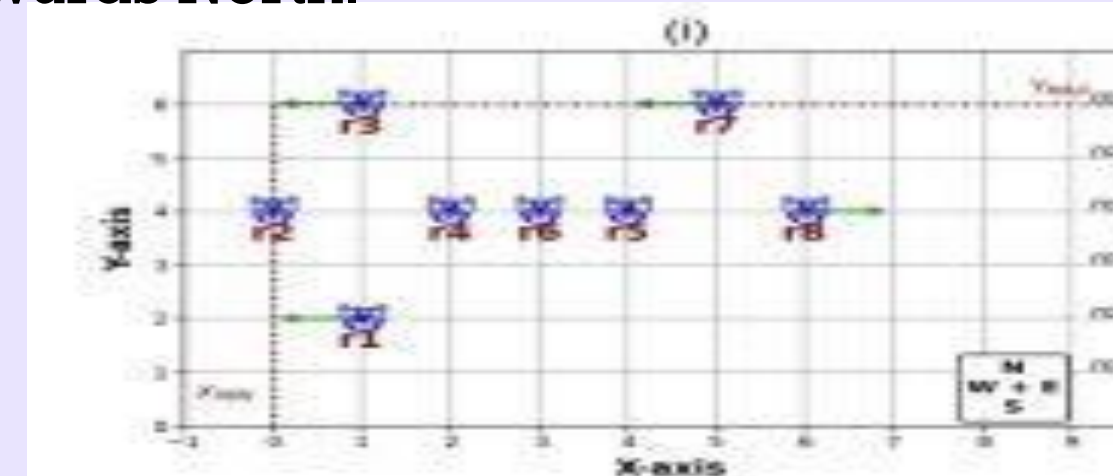
Case ψ_1 : Even Row Within Bounds ($j \leq d$)

- ❖ **Goal:** Move Northward to an odd row.
- ❖ **Action:** Moves to $T_{x,y+1}$ if vacant. If blocked, it sidesteps East ($T_{x+1,y}$) and attempts to move North again; otherwise, it waits.



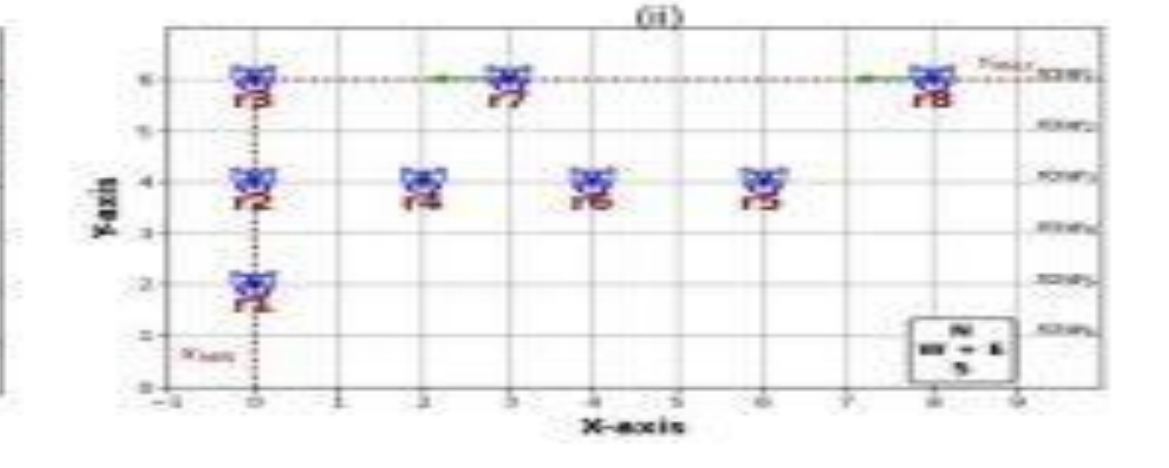
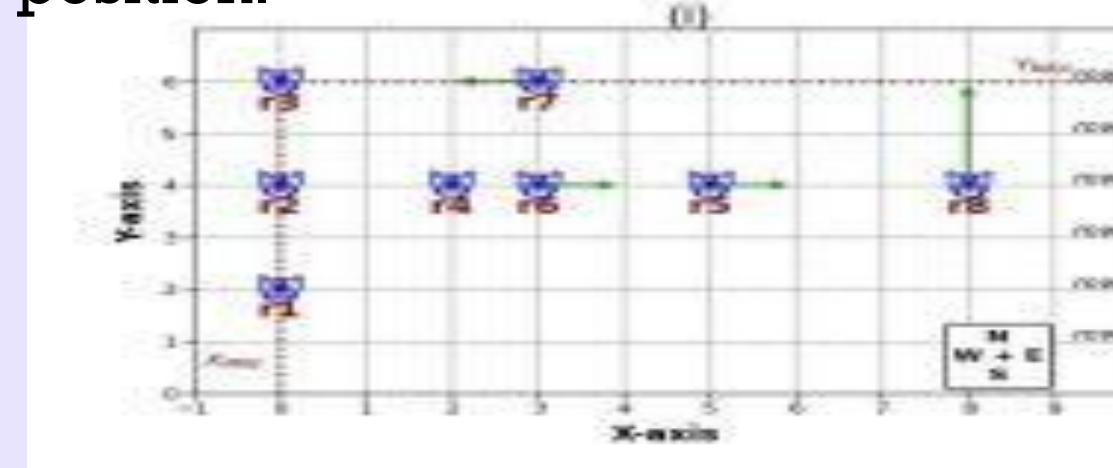
Case ψ_2 : Beyond Distribution Depth ($j > d$)

- ❖ **Goal:** Move Northward to enter the active grid boundary (d^{th} row).
- ❖ **Action:** Advances North to $T_{x,y+1}$ if clear. If blocked, it maneuvers East ($T_{x+1,y}$) to find an open path towards North.



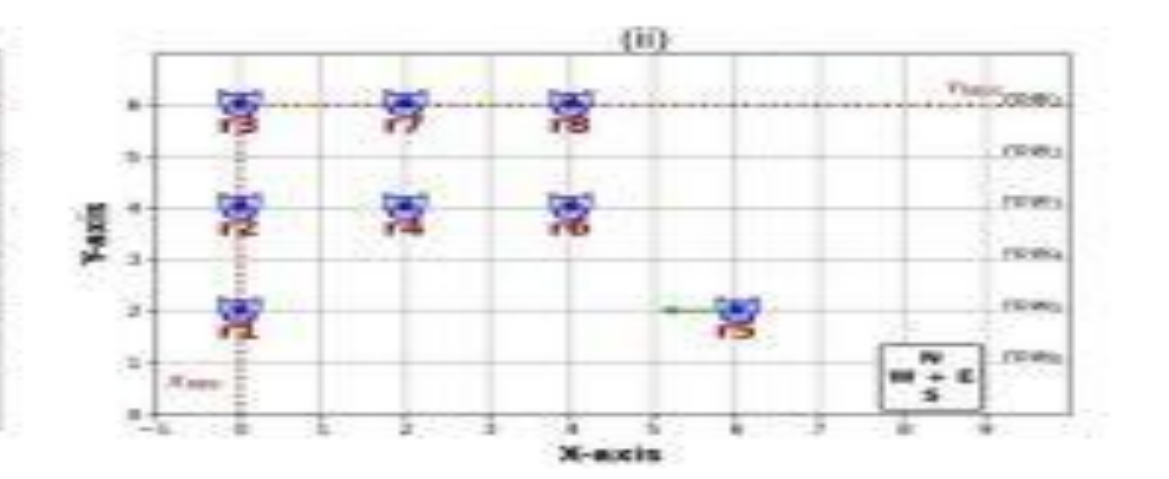
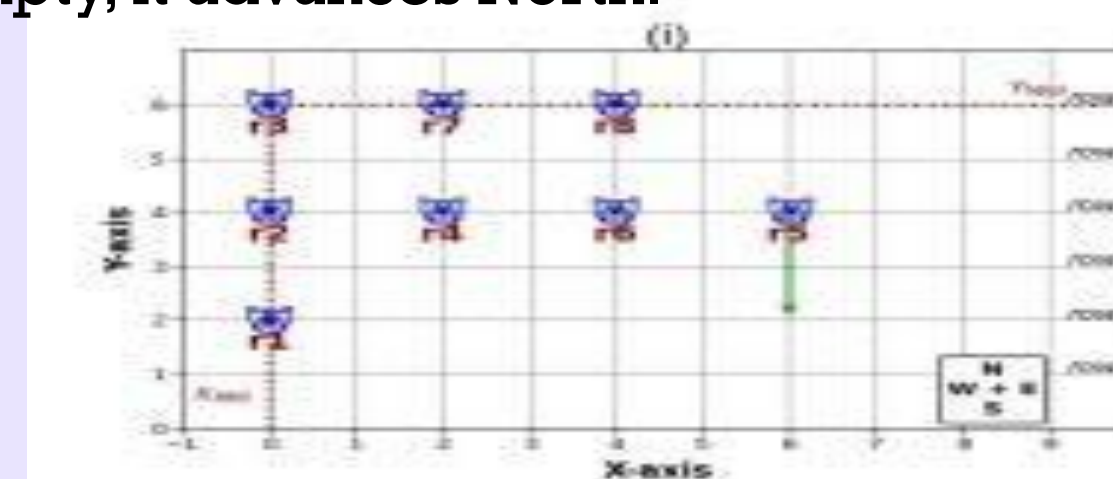
Case ψ_3 : Alignment in Odd Rows

- ❖ **Goal:** Align onto alternate nodes within valid odd rows.
- ❖ **Action:**
 - **Move West:** If a Westward robot has an open target node, it advances West to $T_{x-1,y}$ (if $T_{x-1,y}$ and $T_{x-2,y}$ are clear).
 - **Move East:** If Westward paths are blocked, it shifts East to $T_{x+1,y}$ to secure an alternate node position.



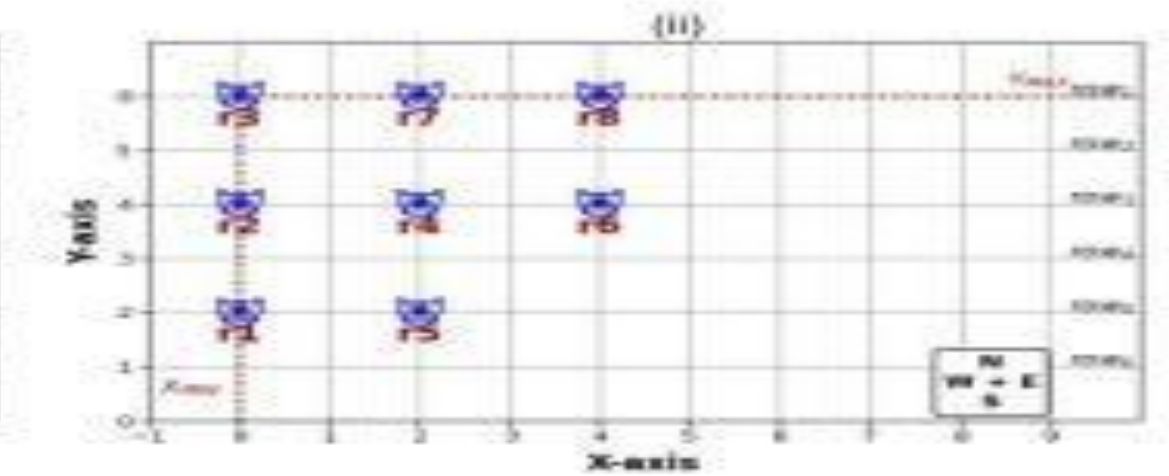
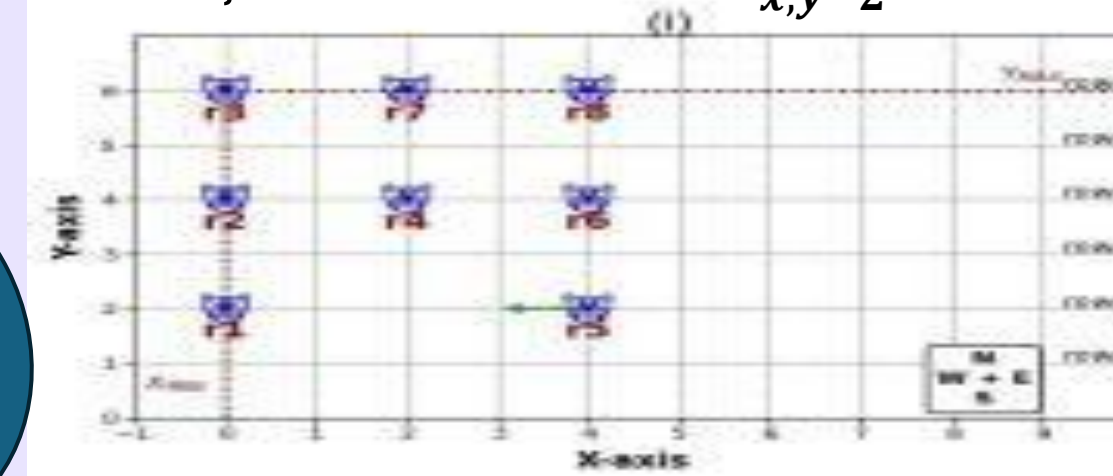
Case ψ_4 : Filling Northward Rows

- ❖ **Goal:** Distribute robots into preceding rows to optimize space.
- ❖ **Action:** If a robot is within the first r_c count of its row and the Northward alternate node $T_{x,y+2}$ is empty, it advances North.



Case ψ_5 : Handling Excess Robots

- ❖ **Goal:** Relocate overflow when a row contains excess robots.
- ❖ **Action:**
 - ❖ **In North-most row:** Shifts Southward to an open odd node $T_{x,y-2}$ and then aligns West.
 - ❖ **In other rows:** Moves North $T_{x,y+2}$ if higher odd rows have fewer than r_c robots. If higher rows are full, it retreats South $T_{x,y-2}$.



Selected References

- **Suzuki & Yamashita (1999):** Pioneered geometric pattern formation for distributed anonymous mobile robots (SIAM JoC).
- **Kshemkalyani & Ali (2019):** Introduced efficient dispersion techniques for mobile robots on graphs (ICDCN).
- **Kshemkalyani, Molla, & Sharma (2020):** Addressed the specific problem of dispersing mobile robots on grids (WALCOM).
- **Barriere et al. (2009):** Explored uniform scattering of autonomous mobile robots specifically within a grid environment (IEEE IPDPS).
- **Bose et al. (2020):** Evaluated arbitrary pattern formation on an infinite grid using asynchronous, oblivious robots (TCS).



TECHNOLOGY TODAY,
A BETTER TOMORROW.

